

LITEON Impact Analysis 2024

LITEON[®]

Foreword

In today's business environment, most companies primarily focus on profitability, often emphasizing financial value while overlooking the multifaceted impacts of their business activities on society, such as environmental and social effects, known as externalities. However, these impacts have a direct or indirect effect on business value and the interests of stakeholders. From a value perspective, there are two main viewpoints: the stakeholder perspective focuses on the positive and negative impacts of business activities on the environment and the resulting social impacts, while the financial perspective concerns how these impacts and dependencies affect the (long-term) financial performance of the business. These two perspectives are essentially interconnected and represent a "double materiality" approach. LITEON aims to be the sustainable best partner for all stakeholders, creating maximum long-term value, both financially and non-financially. Starting in 2022, we recognized this trend and introduced the concept of double materiality in our Sustainability Report. Additionally, we conducted a comprehensive assessment of outcomes and impacts, along with monetization evaluation for key material topics

Methodology

In 2024, LITEON further strengthened its impact monetization calculations, marking our third year of conducting such assessments. In this process, we employed stricter scenarios, particularly in environmental cost calculations, referencing the **U.S. Environmental Protection Agency's (EPA)** 2023 report titled Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances and utilized the **EPA-SC-GHG-Workbook 1.0.2 tool** for calculations. LITEON primarily follows the Value Balancing Alliance (VBA) methodology as the framework for overall ESG metric monetization assessments, using an impact pathway structure. This assessment covers the upstream supply chain, in-house operations, and downstream products and services, analyzing the interdisciplinary interactions between economic, environmental, and social issues.

In the economic dimension, we used the **Gross Value Added (GVA)** methodology to measure the economic value created for stakeholders in our own production and operational processes. Additionally, we applied an **input-output (IO) model** to analyze the external economic value generated by LITEON's procurement activities across the entire industry chain. (**Environmentally Extended Input Output Analysis,**

EEIO) model is also used in calculating externalities in supply chain GHG emission. In the social dimension, we considered academic research papers focusing on the costs of occupational accidents as integral to our assessment framework. In the environmental dimension, we followed the concept of **Environmental Profit and Loss (EP&L)** and referenced tools provided by the **International Foundation for Valuing Impacts (IFVI)**, which is the current organization that includes the research conducted at Harvard Business School's **Impact-Weighted Accounts (IWA)**, to conduct a more comprehensive environmental monetization impact assessment calculation.

Analysis Impact Overview

<i>Item</i>	Aspects	Material Topic Management	Inputs and Outputs in Operational Process	Impacted Stakeholders	SDGs	Methodology	Impact Level (NTD)
<i>U1</i>	Economic	Sustainable Supply Chain Management	Direct Procurement Payment Amount	External Employees	8	VBA, Input-Output Model	522,585,815,236.07
<i>U2</i>	Economic	Supply Chain GHG Emissions	Indirect Procurement Payment Amount	External Employees	8	USEEIO	(87,184,821,600)
<i>C1</i>	Economic	Financial Performance / Taxation	Operating Profit, Depreciation and Amortization, Taxes Paid	Internal and External Stakeholders	8	GVA	60,648,584,000
<i>C2</i>	Social	Occupational Safety and Health	Number of Employees with Improved Health Management	Internal Employees	3	Disability Weight	2,072,140
<i>C3</i>	Social	Occupational Safety and Health	OHS Impact Assessment Input / Employee Health Check and Management Statistics	Internal Employees	3	YLD Method	(233,152.57)
<i>C4</i>	Environmental	Climate and Energy	GHG Emissions (Scope 1 and Scope 2), Self-generated Renewable Energy Usage, Purchased Renewable Energy Usage	Society	7, 13	IFVI	(1,628,626,650)
<i>C5</i>	Environmental, Social	Water Management	Water Resource Consumption	Environment	6	IFVI	(170,691,614.94)
<i>C6</i>	Environmental, Social	Resource Circulation	Waste Disposal	Environment	12	IFVI	(37,467,721.74)

<i>C7</i>	Social	Air Pollutant Emissions	Air Pollutant Emissions	Environment	12	IFVI	(62,709.90)
<i>C8</i>	Social	Community Impact	Community Engagement	Society	3, 13, 17	IFVI	32,550,709.52
<i>D1</i>	Environmental, Economic	Product Innovation Management	Support for Customer Product Success	Customers, Environment	8	IFVI	2,977,852,708.91

U : Upper Stream

C : Corporate

D : Downstream

Impact Description and Results

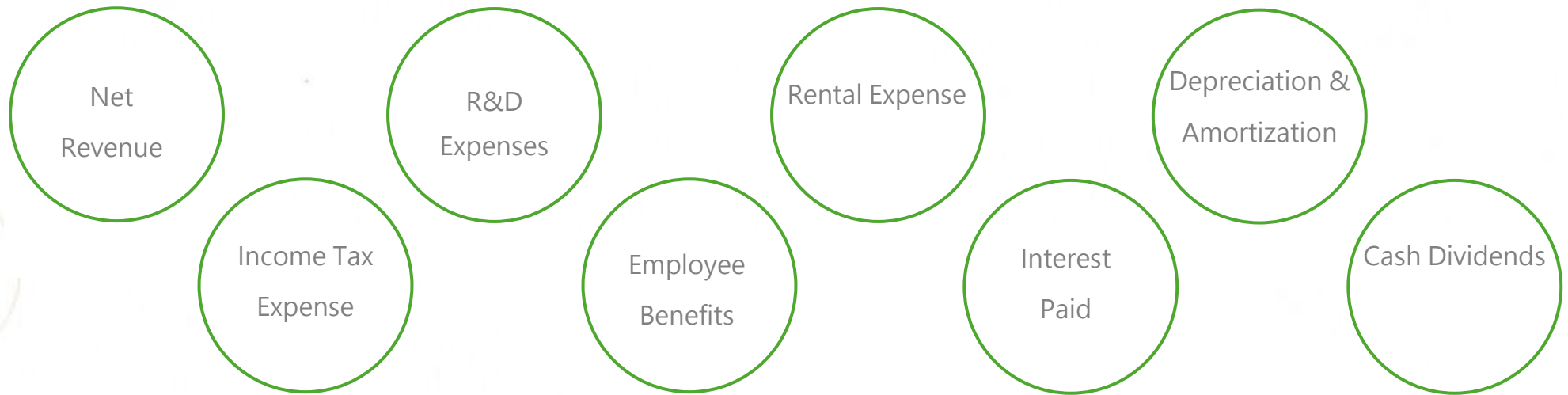
U1 : Sustainable Supply Chain Management

In the supply chain, LITEON has applied input-output model and compiled procurement amount across the upstream industries to calculate total economic benefits created. LITEON's procurement demand is **NTD 70.29 billion**. It resulted in a total economic benefit of **NTD 247.5 billion** and a total labor income of **NTD 27.6**. Based on industry-standard labor coefficients (ILO) and EEIO modeling, this translates to approximately **116,734** jobs globally.

U2 : Supply Chain GHG Emissions

The GHG Emissions is calculated by using the data provided by EPA, stated the emission factors of every industry with NAICS codes and ISIC codes, hence multiply by the social carbon costs provided in IFVI database. The total environmental costs of supply chain is approximately **NTD 87.18 billion**.

C1 : Financial Performance / Tax



In 2024, LITEON generated a direct and positive economic contribution of **275.09 billion** for internal and external stakeholders in the economic dimension calculated using GVA methodology. The aspects include net revenue, taxes, R&D expenses, interest and rental expenses, depreciation and amortization expenses, and cash dividends as well. These contributions create more job opportunities, drive economic growth, yield additional tax revenue for the government, and support beneficial initiatives such as social welfare policies. Through investments in R&D and technological innovation, LITEON promotes overall social technological advancement and industrial transformation.

C2 : Occupational Safety and Health - Number of Employees with Improved Health Management

LITEON has consistently provided sufficient support and care for employee health, offering health checkups as well as activities related to nutrition and exercise to promote physical and mental well-being. Health seminar and health education have been conducted throughout the year, and LITEON continuously monitors year-over-year changes in the number of employees with hypertension, high cholesterol, diabetes, and obesity, and has consistently observed a downward trend overall. The total health improvement benefit from these efforts is **NTD 2.07 million**

C3 : Occupational Safety and Health -Occupational Injuries

In 2024, LITEON employees experienced occupational injuries across multiple operation sites. The total occupational incidents number is 45 and have caused 886 days' workdays to be lost. Although There were no serious injuries, every injury was recorded in detail post-incident training is strictly conducted to avoid similar incidents in the future. By using Years lost due to disability (YLD), Value of Statistical Life (VSL) and Disability Weight calculation method, the estimated social costs are approximately **NTD 0.233 million.**

C4 : Climate and Energy

As an organization in the optoelectronics manufacturing sector, LITEON considers climate adaptation and mitigation, energy management, and carbon emissions reduction as some of the most material issues we face. The social externalities caused by climate change include economic disruptions from extreme weather, ecological damage, public health issues stemming from natural disasters, food and water shortages, and social and political challenges caused by a deteriorating living environment, which ultimately affect social stability. Emissions from corporate activities, such as energy use and direct emissions from chemical processes, have significant consequences.

LITEON records all GHG emissions in detail every year and has consistently reduced carbon emissions year by year. In 2024, with the EPA GHG Workbook model and a 2% discount rate, the environmental external costs of GHG emissions from LITEON corporate operation sites are **NTD 1.63 billion**.

C5 : Water Management

Although LITEON's industry characteristics do not result in significant water consumption, the company continues to record water usage in detail at each site and incorporates both environmental and social costs into its disclosures.

Using the calculation methodology of IFVI, the environmental costs of water consumption of LITEON are around 9% of the overall external environment cost, which is approximately **NTD 165.10 million**. It consists of a few aspects of impacts such as resource costs and ecosystem services as well. On the other hand, In social aspects, the water consumption caused approximately **NTD 5.59 million** considering malnutrition and water-borne disease impacts.

C6 : Waste Disposal

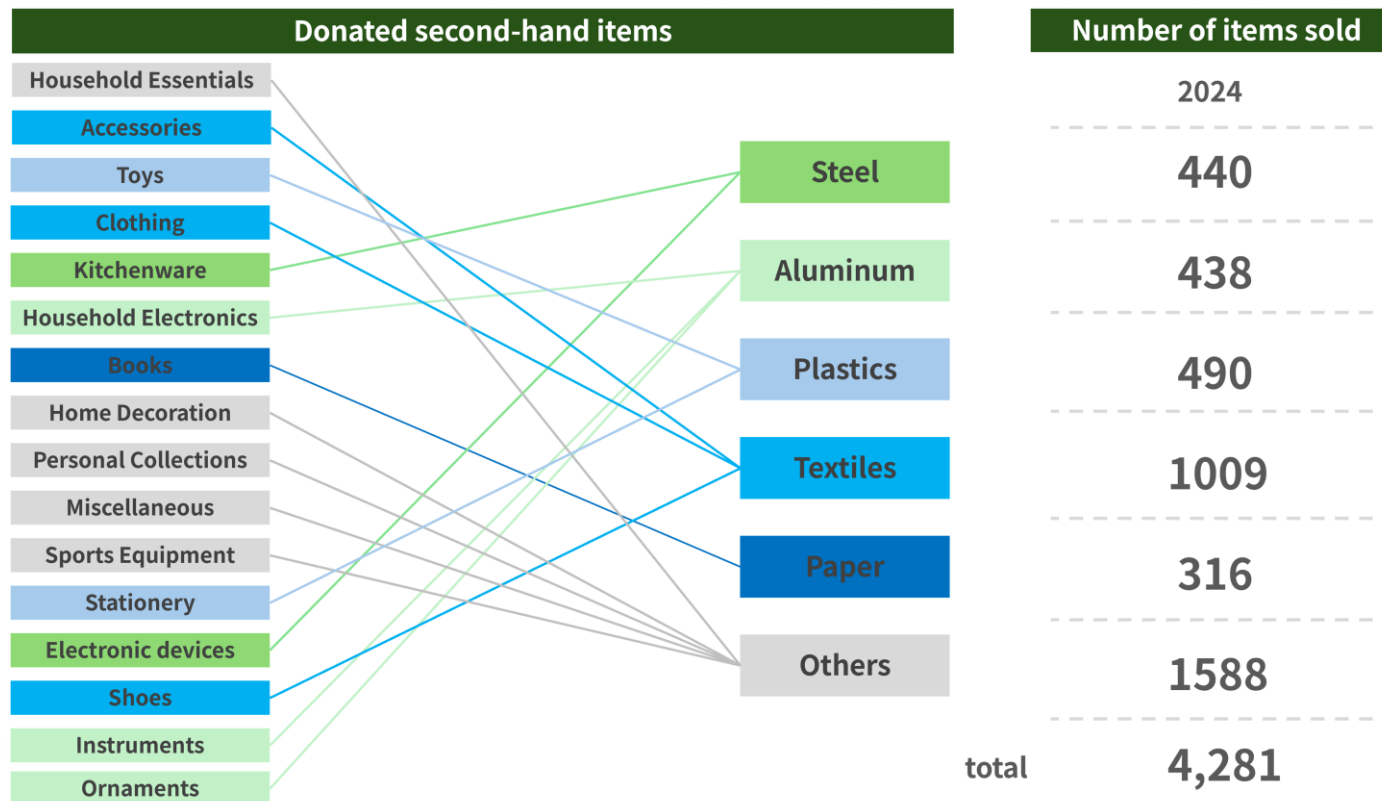
LITEON categorized all company operation sites waste in an organized way by recording hazardous and non-hazardous waste created through landfilling and incineration. The waste consists of manufacturing waste and corporate daily activities waste and LITEON has covered all sites from different countries' operation sites. The external costs are also calculated by using IFVI methodology. The costs of waste disposal of 2024 are **NTD 37.47 million.**

C7 : Air Pollutant

The externalities calculation scope covers VOCs. There are only a few sites that emit VOCs among operating sites. The impacts covered are health, visibility, and agriculture aspects. The total emissions of VOCs are approximately 11 metric tons and the costs are **NTD 62.71 thousand.**

C8 : Community Impact

In 2024, LITEON held a second-hand charity market and invited over 20 corporate partners to join the event. Thousands of participants from corporations and NGOs rallied friends and families to take part, practicing “resource appreciation and green consumption” from fundraising to sales. The objective of the event is to encourage the public to avoid excessive buying. The goods are collected by employees from the household. The benefits of the event are calculated by using IFVI methodology. By selling second-hand goods, the total benefits created are **NTD 32.55 million**.



D1 : Product Innovation Management

LITEON enhanced innovative product design and development on a life cycle thinking approach that incorporates the 3R principle to create environmentally friendly products that are easy to assemble, disassemble, and have low energy consumption. Based on our 2024 shipment volume estimates, our green product design has driven environmental value creation of approximately **NTD 2.98 billion**. The breakdown based on Life Cycle Assessment is also shown below.

Life Cycle Assessment				
Acquisition of Raw Materials	Circular Optimization Design - 581.17 tons CO2e	Sustainable Circular Plastic Use - 8,465.72 tons CO2e		2,135,066.04 USD
Production	Energy Saving Process	Energy Recycling - 599.34 tons CO2e	Automation Design - 3,044.82 tons CO2e	860,021.76 USD
Distribution and Shipping	Logistics Efficiency Management	Product Volume and Materials Reduction - 3.45 tons CO2e	Packaging Design and Recycling - 346.33 tons CO2e	82,548.08 USD
Product Use	Energy Conversion Efficiency - 393,539.05 tons CO2e	Reduce of Electric Components - 9,518.44 tons CO2e	LED packaging technology - 839.51 tons CO2e	95,319,692.00 USD
Disposal and Recycling	system loop optimization and chip integration design reduce the use of electronic components, thus reducing the electronic waste - 34.92 tons CO2e			8,241.12 USD

In 2024, LITEON's external environmental costs were **NTD 1.84 billion**, while green product design initiatives created an estimated **NTD 2.98 billion** in environmental value. By offsetting negative impacts with positive contributions, the net environmental benefit is **NTD 1.14 billion** (net positive).



Going Forward

LITEON continues its sustainability impact assessment this year, with efforts focused on refining methodologies and expanding the scope of social impact calculations. These improvements aim to enhance the relevance of the results to the company's operations. The findings will be incorporated into the material topic identification process and reported to the Corporate Sustainability Committee, supporting informed decision-making. Through ongoing collaboration and the application of technological innovation, LITEON seeks to contribute to a more sustainable future—one that is environmentally responsible, socially inclusive, and economically sound. The company remains committed to balancing financial performance with meaningful non-financial contributions.